

Fatih Maulana

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Engineer and researcher working at the intersection of **distributed systems**, **applied cryptography**, and **AI**. As a **Forward Deployed Engineer**, solves end-to-end business problems with technology and AI — from discovery and solution design through to production delivery — and has shipped production applications for SMEs and early-stage startups since 2024. **First-author** on research in post-quantum cryptography for high-throughput blockchains and in blockchain-based supply chain traceability, the latter awarded a **gold medal** at an international scientific competition. Grant-funded open-source contributor to the **Stellar** ecosystem, building developer tooling for smart contracts, with additional build experience across **Solana** and **Base**. Grounded in security practice: **ISO 27001** and **SOC 2** audit support, penetration testing, and smart contract review.

EDUCATION

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| Aspire Institute (founded by Harvard Business School Faculty)
<i>Aspire Leaders Program</i> | Remote
2026 |
| <ul style="list-style-type: none">Completed an intensive 9-week global leadership program focused on professional development and community impact.Developed and shipped a final capstone project: Tech x Community Solution, addressing local socio-economic challenges through digital infrastructure. | |
| Universiti Utara Malaysia (UUM)
<i>Bachelor of Computer Science with Honors (Exchange Program)</i> | Kedah, Malaysia
2025 – 2026 |
| <ul style="list-style-type: none">Focus: Artificial Intelligence, Software Development Management, Cybersecurity, Blockchain, IT Resource Management. | |
| UIN Sunan Gunung Djati
<i>Bachelor of Science in Informatics Engineering</i> | Bandung, Indonesia
2023 – 2027 |
| <ul style="list-style-type: none">GPA: 3.84 / 4.00 (Summa Cum Laude Track)Focus: Distributed Computing, Cybersecurity, Blockchain, and Networks. | |
| Private Boarding School Minhajul Haq
<i>Senior High School Diploma</i> | Purwakarta, Indonesia
2020 – 2023 |
| <ul style="list-style-type: none">Average Final Score: 95.00. Focused on Language Proficiency, Theology, and Leadership. | |

WORK EXPERIENCE

- | | |
|--|--|
| Stellar Development Foundation (Stellar Community Fund)
<i>Instaward Awardee — Developer Tooling</i> | Remote
2026 |
| <ul style="list-style-type: none">Awarded a 4,800 USD in XLM grant through the Stellar Community Fund Instaward to build Evergreen, an open-source developer tool for Soroban smart contracts.Evergreen automates TTL (time-to-live) and state-archival management in Soroban, helping teams keep contract state alive and avoid unexpected archival in production. | |
| BlockDev.ID
<i>City Builders Lead</i> | Bandung
June – July 2026 |
| <ul style="list-style-type: none">Spearheaded a 1-month intensive Web3 incubator in Bandung, mentoring 15 developers in building end-to-end DeFi MVPs.Designed and delivered a hands-on technical curriculum on EVM development, focusing on Solidity, Foundry, and Wagmi/Next.js integration on the Base L2 network.Guided engineering teams through the entire product lifecycle, from system architecture and tokenomics design to Go-To-Market strategy and hackathon pitches. | |
| Daemon Protocol
<i>Forward Deployed Engineer & Product Owner</i> | Remote
Apr 2026 – Present |
| <ul style="list-style-type: none">Forward Deployment: Embedded directly with clients to solve end-to-end business problems with technology and AI — owning discovery, solution design, implementation, and production delivery.Product Leadership & Engineering: Spearheading the development of Ares, an agentic AI platform utilizing the Hermes workflow for automated vulnerability detection in Solana smart contracts.AI Optimization: Fine-tuning LLMs on authentic malware, vulnerabilities and security datasets to ensure high true-positive rates and minimize false positives during complex security audits.Technical Research: Leading research and authoring a scientific paper evaluating AI agent performance in identifying critical security flaws within the Solana Sealevel runtime. | |
| Superteam Campus Club UIN Bandung
<i>Lead</i> | Bandung, Indonesia
Feb 2026 – Present |

- Leading the official Solana-focused blockchain community on campus, managing internal operations and ecosystem growth.
- Facilitating the shipping of real-world Web3 projects and coordinating participation in global Superteam/Solana events.
- Spearheading developer relations and awareness campaigns to onboard students into the Solana ecosystem.

Wo-Men in Tech Security

Remote

GRC Member

Nov 2025 – March 2026

- Assisted in audit preparation for ISO 27001 and SOC 2 by gathering and tagging technical evidence from IT teams.
- Monitored GRC automation platforms (Vanta, Drata) to identify and remediate non-compliant devices, achieving 100% policy adherence.
- Administered security awareness training with a 95%+ completion rate and maintained the central Risk Register.

Changlun Smart Agriculture System

Malaysia

Full-Stack Engineer (Academic Collaboration) with Dr.Rohaya @UUM

Dec 2025

- Architected a scalable SaaS platform for agricultural data management using **Django (Python)**, focusing on backend logic and database schema optimization.
- Designed RESTful APIs to handle data ingestion from IoT sensors, ensuring data integrity and efficient query performance.
- Collaborated directly with research professors to translate agricultural domain requirements into technical system specifications.

National Student Election System for CSSMoRA Indonesia

Remote

Security Consultant & Interim Lead

Nov – Dec 2025

- Salvaged a critical national voting project by stepping in as Lead Developer just **24 hours prior to launch**; strategically pivoted from a failed complex web architecture to a serverless Google Apps Script solution to guarantee on-time delivery.
- Engineered a **token-based validation mechanism** within the script backend to enforce strict unique-voter constraints, successfully processing votes with zero fraud incidents despite platform limitations.

Ibunda.id

Bandung, Indonesia

Technical Writer & Analyst

Jun 2025 – Oct 2025

- Developed detailed system documentation and role-specific user manuals for three core products using GitLab and Notion, reducing onboarding time.
- Conducted technical presentations on user sentiment analysis utilizing **Support Vector Machine (SVM)** algorithms, translating complex AI concepts into actionable insights for stakeholders.
- Documented API contracts and service boundaries within a microservices architecture to ensure clarity in inter-service communication.
- Monitored code quality via **SonarQube** and analyzed application logs in **Grafana** to proactively identify bug patterns.

StudentxCEOs Bandung Chapter

Bandung, Indonesia

Front-End Engineer

Jan 2025 – Jun 2025

- Developed UI components for six web pages including login and registration using **Next.js** and **TypeScript**.
- Resolved server-side rendering issues in Next.js that caused page load delays, optimizing component rendering for better performance.
- Collaborated with UI/UX designers and backend teams to ensure seamless integration and consistent user experience.

Focustudio.online

Remote

Founder & Developer

Oct 2024 – Present

- Founded a personal web agency, managing client projects from development to deployment.
- Implemented **CI/CD pipelines** using Cloudflare Tunnel and GitHub Actions to automate deployment workflows.
- Ensured site reliability and continuous delivery through rigorous automation practices.

Indonesian Ministry of Religious Affairs (w/ LPDP)

Jakarta, Indonesia

Technical Support Specialist & Virtual Coordinator

Apr 2024 – Apr 2025

- Managed end-to-end virtual infrastructure for national-level PBSB scholarship selection, facilitating interviews for hundreds of candidates.
- Orchestrated complex Zoom logistics including breakout rooms and secure access controls for government officials.

QUANTITATIVE & SCIENTIFIC RESEARCH EXPERIENCE

- Post-Quantum Cryptography for Solana Migration** | *CRYSTALS-Dilithium (ML-DSA-65), Rust, Rayon, Python* 2026
- **First-author feasibility study** benchmarking **CRYSTALS-Dilithium (ML-DSA-65)** against Ed25519 for a Solana-compatible migration, framed around the **Sealevel/SigVerify** pipeline; submitted to **Blockchain Kaigi 2026** (Tokyo).
 - Measured median verification latency of **281.7 ± 26.9 μs** (Dilithium3) against **51.6 ± 3.5 μs** (Ed25519) over three reproducible runs; Mann–Whitney and Welch tests reject equal distributions ($d \approx 1.10$).
 - Mapped a **Rayon** thread pool to SigVerify-style batching (64 signatures, 8 threads) for a 17.5% relative throughput index, and simulated MTU fragmentation of 3,309-byte signatures into three 1280-byte datagrams (97.3% delivery at 1% independent loss; 96.0% under correlated loss).
 - Built an offline analytical **compute-unit admission model** predicting stall thresholds near concurrency 64 under the 1.4M CU cap, plus an off-chain Dilithium3 vault prototype with audit logging.
- HalalChain: Trustless Batch Traceability on Base L2** | *Solidity, Base Network, IPFS, Hardhat, Next.js* 2026
- **First-author paper** on a decentralized halal batch-traceability protocol deployed on **Base** (Ethereum L2), anchoring certification status, IPFS content identifiers, and audit metadata on a public ledger while keeping certificates, lab reports, and ingredient lists off-chain.
 - Designed a three-role smart contract (Producer / Auditor / Consumer) with **OpenZeppelin RBAC**, encoding auditor rejection and revision history as first-class protocol states rather than off-ledger corrections.
 - Benchmarked on Ethereum Sepolia: **registerBatch** at **155,506 gas** and **verifyBatch** at **79,214 gas**, showing affordability for Indonesian SME producers under BPJPH/MUI halal labelling regulation.
 - Shipped an open-source implementation — Solidity contracts, Hardhat test suite, and a bilingual producer/auditor/consumer dashboard with **wallet-free QR verification** for end consumers.
- High-Energy Physics & Quantum Computing Researcher** | *Independent* Dec 2025
- Designed and benchmarked hybrid quantum-classical algorithms to classify High-Energy Physics (HEP) events using the **ATLAS Higgs Boson Challenge dataset** ($N = 800$ events).
 - Implemented a dimensionality reduction pipeline using **PCA** to map 30 physical features into NISQ-compatible latent spaces (4 and 8 qubits).
 - Engineered a **Variational Quantum Circuit** utilizing **ZZFeatureMap** and **RealAmplitudes** ansatz.
 - **Key Finding:** Demonstrated that increasing circuit depth ($d = 2$) yielded a **56.2% accuracy**, outperforming wider 8-qubit circuits (50.6%) which suffered from the **Barren Plateau problem**.
 - Optimized parameters utilizing the **COBYLA** gradient-free optimizer via **Qiskit SDK**.
- Quantitative Analyst & Machine Learning Researcher** | *UUM* Late 2025
- Conducted a rigorous statistical evaluation of technical indicators (**RSI, SMA**) to test the weak-form **Efficient Market Hypothesis (EMH)** on Bitcoin (BTC/USD) data.
 - Developed a backtesting engine in **Python (Pandas, NumPy)** to engineer momentum features and volatility metrics.
 - Evaluated k-NN, Decision Trees, and Logistic Regression; utilized **ROC/AUC analysis** to demonstrate that standard technical indicators failed to achieve statistical significance ($AUC \approx 0.5$) due to market efficiency.
 - Diagnosed model failure modes (overfitting vs. mode collapse) proving the stochastic nature of short-term price movements.
- Statistical Analysis of NYSE Market Trends** | *Python, Pandas, Matplotlib*
- Conducted rigorous Exploratory Data Analysis (EDA) on NYSE historical data to identify long-term trend components and seasonality.
 - Visualized volatility regimes and price distributions utilizing Matplotlib/Seaborn, providing insights into market stability periods.
 - Implemented efficient data pipelines to clean and process raw financial datasets, ensuring data integrity for downstream analysis.
- Blockchain Market Researcher (Contract)** | *Toobit Exchange* Oct 2025 – Dec 2025
- Directly commissioned by the ASEAN Marketing Manager to execute a comprehensive strategic analysis of the Malaysian blockchain and cryptocurrency landscape.
 - Conducted due diligence on local regulatory frameworks and competitor market share, providing data-driven recommendations for regional expansion strategies.
 - Analyzed user demographic trends and crypto-adoption vectors to formulate actionable insights for optimizing local community engagement and marketing campaigns.

TECHNICAL SKILLS

Quantitative & Scientific: Python (NumPy, Pandas, SciPy, Scikit-learn), Qiskit (Quantum), MATLAB, R, Statistics, Financial Modeling

Languages: Python, C, C++, Java, SQL, Bash, JavaScript (TypeScript), PHP, Dart

Software Engineering: Django, Next.js, Flutter, Spring Boot, OOP, System Design, Microservices, CI/CD (GitHub Actions)

Security & Infrastructure: Kali Linux, Burp Suite, OWASP ZAP, Wireshark, Docker, Cloudflare, SonarQube, Grafana

Tools: Git, LaTeX, Jupyter Notebooks, Postman, Figma, Notion, Trello, Jira

Languages: English (Professional), Indonesian (Native), Arabic (Professional), Melayu (Conversational), Japanese (Conversational)

Interests: Classical Piano, Strategic Games (Chess/Poker), Financial Markets, Open Source Software

SELECTED TECHNICAL PROJECTS

- Nemesis Protocol** | *Founding Engineer & Business Development — Solana, DePIN/IoT* 2026
- Building **financing rails for Indonesia's EV ecosystem**, combining on-chain settlement with IoT/DePIN vehicle data.
 - Owning the commercial side alongside engineering: cost structure and unit-economics mapping, runway planning, and investor/competition pitch materials.
- Auditorum: Web3 Security Ideation** | *Blockchain Auditor Concept* Feb 2026
- Architected a Web3 auditing framework designed to bridge the trust gap between blockchain startups and institutional investors.
 - Designed risk assessment models to evaluate smart contract vulnerabilities and governance transparency.
- Freelance DeFi Stability Platform** | *Solana, Rust, Stablecoins* Jan 2026
- Developed a DeFi platform prototype utilizing stablecoins and staking mechanisms to provide financial stability for freelancers.
 - Integrated a "micro-task" engine allowing users to increase collateral/income through simple verifiable on-chain jobs.
- E-Commerce Website Penetration Testing** | *Security Tester* Jan 2025
- Conducted comprehensive penetration testing using Burp Suite and OWASP ZAP; identified vulnerabilities including XSS, IDOR, and broken access control.
 - Assessed severity using CVSS scoring and generated detailed remediation reports for stakeholders.
- Dijkstra Pathfinding Algorithm Visualizer** | *Java, Graph Theory* Jan 2024
- Developed a high-performance GUI application to visualize Dijkstra's shortest-path algorithm on weighted graphs.
 - Implemented optimal data structures (**Priority Queue**) to ensure $O(E + V \log V)$ time complexity.
- Embedded IoT Item Counter System** | *C Language, Arduino* Jan 2024
- Engineered a low-level embedded system using **C** to process real-time sensor data from ultrasonic modules.
 - Implemented direct memory management and hardware interrupt handling to minimize latency.
- DIY Brain-Computer Interface (BCI)** | *C++, ESP32, Signal Processing* Dec 2025
- Engineered a low-cost BCI prototype using ESP32 microcontroller to capture and process bio-signals (EEG) in real-time.
 - Implemented digital filters (Band-pass, Notch) in C++ to isolate neural activity from electrical noise, mirroring signal extraction techniques.
- Financial Transaction & Banking Simulation** | *Java, OOP* Jan 2023
- Built a banking simulation app with core features (deposit, withdrawal, investment) using strict OOP principles.
 - Implemented backend logic for ROI calculation and transaction integrity.
- Other Engineering Projects** | *Various Technologies* 2023 – 2025
- **StudentxCEOs Website:** Developed landing page with login/registration using Next.js and TypeScript.
 - **Mobile E-commerce Prototype:** Designed functional prototype with Flutter (Dart) focusing on state management.
 - **Campus Whistle-blowing System:** Built a CRUD-based reporting system using PHP and MySQL.
 - **Psychology-based Game:** Led a team to build a game prototype using Unity and C#.
 - **Library Management System:** Developed a Java desktop app for managing book loans and members.

LEADERSHIP, AWARDS & ACTIVITIES

- Gold Medal — ICYMS** | *International Competition of Young Moslem Scientists* 2026
- Won gold with **HalalChain**, a blockchain-based halal traceability system, presented as both a product concept and a research paper.
- Web3 Speaker & Ecosystem Contributor** | *Indonesia* 2025 – Present
- Speaker at **6+ Web3 events** across Indonesia with Indodax, ID.NFT, Bybit Indonesia, Bitget, Superteam Indonesia, Stellar Indonesia, and DevWeb3 Bandung.
 - Directly supporting Web3 ecosystem growth in Indonesia through developer onboarding, technical education, and community building.
- Head of Financial Literacy Dept** | *CSSMORA UIN SGD* Jul 2025 – Present
- Spearheaded the design of a comprehensive financial literacy curriculum, mentoring members on **Fundamental & Technical Analysis**.
 - Led a cross-functional team to execute educational workshops, bridging theoretical finance concepts with practical application.
- Academic Board Member (NALAR)** | *HIMATIF UIN SGD* Mar 2025 – Present
- Mentored peers on core Computer Science concepts (Algorithms, Calculus), facilitating study groups to enhance academic performance.
 - Served as a technical speaker for departmental events, translating complex curriculum topics into accessible modules.

Google Developer Student Club (GDSC) | *Member*

Jan 2023 – Jan 2024

- Applied Machine Learning techniques including Classification, Regression, and Clustering using Python/NumPy.
- Contributed to open-source repositories related to Flutter and Next.js.

Honors & Additional Leadership

- **Speaker**, Youth Today x Schoters at JoinAIESEC Event (Youth Empowerment).
- **Best Speaker**, English Domestic Competition (Pare).
- **3rd Place**, Chemical Engineering at Madrasah Science Competition (West Java).
- **Delegates**, AIESEC Future Leaders Winter Peak.
- **Head of Marketing Bureau**, CSSMORA UIN SGD (Economic Development Division).
- **Secretary of Art Division**, OSMHA (Minhajul Haq Student Council).

CERTIFICATIONS

Cisco Certified Network Associate (CCNAv7) - In Progress (Focus on Switching, Routing, and Wireless Essentials)

Cybersecurity Introduction - Codingstudio.id (Jan 2025 – Sep 2027)

CS50's Introduction to Computer Science - Harvard/edX (Dec 2024 – Dec 2025)

GRC Introduction - Codingstudio.id (Dec 2024 – Dec 2025)

Website Penetration Testing - Codelamp Bootcamp Indonesia (Nov 2025 – Nov 2026)